04

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING
01 APR 2025
EXAMINATIONS & REGISTRATION

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# University of Colombo, Sri Lanka

*University of Colombo School of Computing*

## BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

First Year Examination - Semester II - UCSC AY21 [held in March/April 2025]

## IS1210 Database Systems

(Two (2) Hours)

**Answer ALL questions**

Number of Pages = 11 Number of Questions = 3

To be completed by the candidate

Index Number: | | | | | | | | |

**Important Instructions to Candidates:**

I. Students should answer in the medium of English language only using the space provided in this question paper.

II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.

III. Write your index number CLEARLY on each and every page of this Question paper.

IV. This paper consists of **3** questions in **11** pages (including the Cover Page).

V. **Answer ALL questions.**

VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones **are not allowed.**

VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

**To be completed by the examiners**

| Question No | Marks |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| Total | |

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**Question 1 – Total 40 Marks**

Elite Events Solutions (EES) is a leading event management company specialized in organizing and executing a wide range of events for diverse clients. To enhance its operations, the company requires a robust database system to manage its core components.

Each event is uniquely identified by an event ID and includes key details such as its name, date, location, and budget. Events are further categorized into corporate, personal, and community types. Corporate events have additional details, such as the sponsoring organization and the expected number of attendees. Personal events record the host's name and guest count, while community events are defined as those without a specific sponsor or individual host. The system also accommodates events that may not fall into these predefined categories, allowing for flexibility to address the company's diverse offerings.

EES serves two types of clients, such as individuals and organizations. Each client is identified by a unique client ID and includes basic information such as name, multiple phone numbers, and an email address. Clients can be either individuals, identified by a personal ID and birthdate, or organizations, identified by a registration number and a contact person. A client cannot simultaneously act as both an individual and an organization. A client can manage multiple events, but each event is exclusively linked to one client.

The EES company's staff plays a vital role in ensuring successful event execution. Each staff member is identified by a unique staff ID and information such as name, role, and phone number. Staff members may be assigned to multiple events, and each event typically requires contributions from several staff members. In addition, staff members are responsible for managing specific services related to an event.

Staff members follow a hierarchical structure, where some staff members supervise others. Each staff member may be supervised by one senior staff member, and a senior staff member may oversee multiple subordinates.

EES offers a variety of services, such as catering, decoration, and transportation, each uniquely identified by a service ID and described by its type and cost. Events may require multiple services, and each service can be used across multiple events.

(a) Draw an EER diagram showing the necessary entities, attributes along with the primary key, and relationships for the above scenario. Specify the cardinality and participation constraints for each relationship set. Clearly state all your assumptions, if any.

**[20 Marks]**

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(b) Map the EER diagram developed in (a) above to a relational data model. Underline the primary key of each relation and explicitly indicate any foreign keys by marking them with (FK).

**[20 Marks]**

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**Question 2 – Total 30 Marks**

(a). For each of the following statements, determine whether it is True or False. Regardless of the answer, you must provide a proper justification to explain why the statement is true or false.

**[2 X 10 = 20 Marks]**

1) If A → B and B → C, then A → C always holds in a relational schema.

2) Functional dependencies apply only to primary keys.

3) If A → B and A → C, then A → BC.

4) A table in First Normal Form (1NF) must not contain repeating groups.

5) A relation in Second Normal Form (2NF) can have partial dependencies.

6) Normalization helps in removing redundancy completely and improving data integrity.

7) Deletion anomalies occur when deleting a record's field value, essential data gets deleted.

8) Modification anomalies occur when updating a record, does not affect all instances of that data in a table.

9) Reducing update anomalies requires the denormalization of tables.

10) The main goal of normalization is query optimization.

(b). A company tracks employee assignments to projects. Each employee can work on multiple projects, and each project has a manager. Convert the table into the third Normal Form (3NF). Indicate the steps, and tables in each step with the keys and any assumptions. **[10 Marks]**

**Employee_Project**

| Employee_ID | Employee_ Name | Project_ID | Project_Name | Manager_ID | Manager_Name | Manager_Email |
|---|---|---|---|---|---|---|
| 201 | John | P101 | AI Research | M1 | Alice | alice@xyz.com |
| 202 | Mike | P102 | Data Mining | M2 | Bob | bob@xyz.com |
| 201 | John | P102 | Data Mining | M2 | Bob | bob@xyz.com |
| 203 | Sarah | P101 | AI Research | M1 | Alice | alice@xyz.com |

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## Question 3 – Total 30 Marks

A hospital maintains a database to manage Doctors, Patients and their appointments. The following tables provide details about these entities. Based on the tables given, write SQL statements for the questions given below.

**Doctor Table:** Stores details about hospital's doctors.

| Column Name | Data Type | Description |
|---|---|---|
| doctorID | INT | Unique ID for each doctor, auto-incremented. |
| name | VARCHAR(50) | Doctor's name, Cannot be null. |
| specialization | VARCHAR(50) | A doctor's field of expertise, Cannot be null. |
| experience | INT | Years of experience, Cannot be null. |

**Patient Table:** Stores information about the hospital's patients.

| Column Name | Data Type | Description |
|---|---|---|
| patientID | INT | Unique ID for each patient, auto-incremented. |
| name | VARCHAR(50) | The patient's name, Cannot be null. |
| age | INT | Patient's age, Cannot be null. |
| disease | VARCHAR(100) | Disease diagnosis, Cannot be null. |

**Appointment Table:** Keeps track of doctor-patient appointments.

| Column Name | Data Type | Description |
|---|---|---|
| appointmentID | INT | Unique ID for each appointment, auto-incremented. |
| patientID | INT | Foreign key referencing Patients(patientID), Cannot be null, If a patient is deleted, their appointments should also be deleted. |
| doctorID | INT | Foreign key referencing Doctors(doctorID), Cannot be null, If a doctor is deleted, their appointments should also be deleted. |
| appointmentDate | DATE | The date of the appointment, Cannot be null |

1) Write an SQL statement to create the **Appointment** table. **[6 Marks]**

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2) Write an SQL statement to insert the following appointment record into the appointment table. "Alice Brown" with **patientID** = 1, "Dr. Williams" with **doctorID** = 1, Date - 2024-03-01.
**[3 Marks]**

3) Increase "Dr. Williams's" experience by 2 years. **[3 Marks]**

4) Remove "Alice Brown's" appointment with "Dr. Williams" on '2024-03-01'. **[2 Marks]**

5) Retrieve patient names, doctor names, and appointment dates in a single query by joining the necessary tables. **[3 Marks]**

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6) Display the name and age of the oldest patient. **[2 Marks]**

7) Write an SQL statement to create a view named **ExperiencedDoctors** that contains doctors with more than 10 years of experience. Ensure that any insert/update operations on this view do not violate this condition. **[4 Marks]**

8) Write an SQL query to count the number of patients diagnosed with each disease, then order the results in descending order of patient count. **[2 Marks]**

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9) Write an SQL query to find the average experience of doctors for each specialization and order the results from highest to lowest experience. **[2 Marks]**

10) Write an SQL query to retrieve a combined list of all people (doctors and patients) in the hospital. The result should include their names and specify whether they are a Doctor or a Patient. **[3 Marks]**

*******The End*******